OPERATING SYSTEMS – ASSIGNMENT 1

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TASK 12 – Re-evaluate and Adapt the Design (8 Marks)

Changed Requirement: The hospital adds remote monitoring devices in patients' homes that send readings over the internet and moves patient records to a cloud service shared with partner hospitals. Re-evaluate the original design; RETAIN, MODIFY or REDESIGN.

A. Impact Analysis – 2 Marks: Identify affected OS types, services and protection mechanisms.

B. Evaluation of Original Design – 2 Marks: Evaluate response time, reliability and security.

C. Adaptation or Redesign – 2 Marks: Describe the revised plan, e.g. embedded RTOS on home devices and distributed system for shared records.

D. Evidence-Based Defence – 2 Marks: Defend the decision using evidence and trade-offs.

Evidence Expected: Clear impact analysis, technically supported evaluation, appropriate design response and evidence-based defence.

ANSWER

A. Impact Analysis – 2 Marks

The new requirement adds home patient-monitoring devices that send readings over the public internet and moves patient records to a cloud service shared with partner hospitals. The original local RTOS monitors, mobile tablets, desktop workstations, database cluster and distributed hospital link remain relevant, but the architecture must be extended.

OS types affected: home devices need an embedded OS, preferably a small RTOS where alarms or local safety actions have deadlines. Cloud records require a distributed, virtualised server environment with high availability and elastic capacity.

Services affected: secure network communication, remote device management, authentication, time synchronisation, logging, remote update, cloud storage, replication, backup and API services become more important.

Protection affected: device identity and certificate lifecycle, mutual TLS, encryption in transit and at rest, secure boot, signed firmware, least privilege, MFA, role-based access, audit, network segmentation and privacy controls are required.

Remote connectivity introduces variable internet latency, packet loss, outages, device theft and a larger attack surface. These conditions must be treated as normal failure cases.

B. Evaluation of the Original Design – 2 Marks

The original design is a strong foundation but cannot be retained unchanged. Local hard real-time monitoring remains appropriate because it protects patients when the internet is slow or unavailable. A hospital-only cluster is not sufficient as the complete record-sharing solution once records are hosted in the cloud and accessed by three hospitals and home devices.

Response time: internet latency is variable and cannot provide a deterministic 10 ms alarm path. Therefore, home devices must detect urgent conditions locally and raise a local alert where applicable; cloud transmission is for clinician notification, trend analysis and record synchronisation, not the only alarm mechanism. Reliability: devices need store-and-forward queues, retry with backoff, local timestamps and safe behaviour during disconnection. Cloud services need multiple availability zones/regions, backups and tested disaster recovery. Security: public internet access requires stronger device identity, encryption, API authorisation, monitoring and lifecycle management.

C. Adaptation / Revised Design – 2 Marks

D. Evidence-Based Defence – 2 Marks

Decision: MODIFY the original design rather than fully retain or completely redesign it. The local hard RTOS monitor architecture is retained because a 10 ms safety response cannot be guaranteed over the public internet. The database architecture is modified to a distributed cloud records service with replication and strong identity/access controls. Home devices gain embedded security, local detection and store-and-forward communication.

The evidence supports this choice: the 10 ms requirement needs bounded local processing, while internet and cloud communication have variable latency. Clustering and replication improve availability but do not remove network partitions. Encryption, device identity and audit address the expanded attack surface. The trade-off is greater operational complexity, cloud/network cost and security governance effort, in exchange for remote monitoring, cross-hospital access and scalability. A tested offline mode and local alarms are mandatory safeguards.

Submission Checklist

Include all C source files, scripts, configuration files, outputs and genuine screenshots in the repository.

Run the programs on Linux and insert actual strace/time/process/thread measurements.

Create a README.md with compiler version, Linux distribution/version, compile commands and run instructions.

Commit progressively, with at least one meaningful commit for each task.

Layer | Revised design

Home monitoring device | Embedded RTOS or secure embedded Linux depending on timing and hardware; local threshold detection, local alert, watchdog, secure boot, signed OTA updates and encrypted local queue.

Connectivity | Mutual TLS device authentication, encrypted transport, API gateway, rate limiting, device certificates, retry/backoff and store-and-forward during outages.

Cloud records | Distributed cloud database/service with replication, backups, access control, audit and disaster recovery. Separate tenant/hospital access policies and interoperable APIs.

Hospital systems | Local clinical systems and database cache/replica for continuity; secure synchronisation with cloud; local alarm network isolated from cloud dependency.

Clinicians | Managed tablets and desktops using MFA, role-based access and audit. Remote access through secure identity-aware access controls.